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AWS RDS

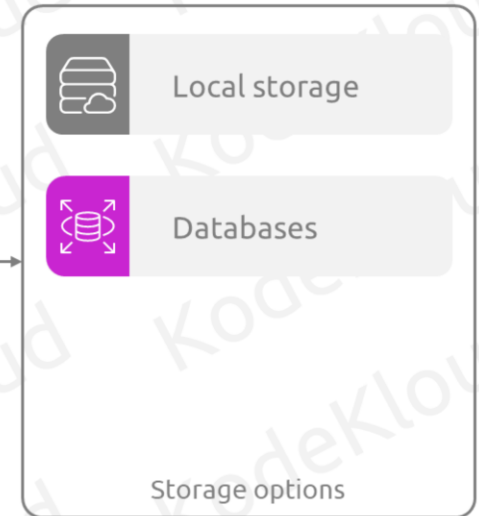
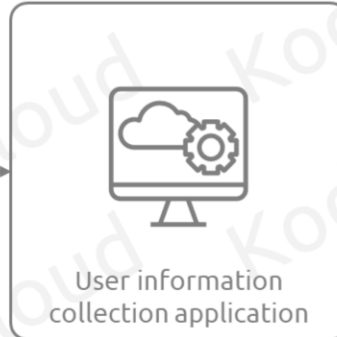
Understanding the need for databases



User 01



User 02



On Premises Data Centre

On-Premises data center



MySQL



PostgreSQL



MariaDB

ORACLE

Oracle



Microsoft SQL
Server

Overview of RDS and its benefits



High agility and adaptability



Lower risk of hardware failures and downtime



Easier managing disaster recovery capabilities



Easier maintaining security and compliance standards



Reduce dependency on internal IT resources for database management and support



Overview of AWS Console



Spinning up MariaDB RDS Instance



Different Database Offerings in AWS RDS



MySQL



PostgreSQL



MariaDB

ORACLE

Oracle

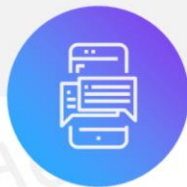


Microsoft SQL Server

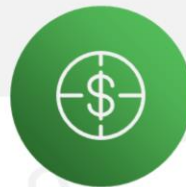
Each RDS version offers seamless integration with other AWS services and tools



Suits all different application use cases for an organization



Focus on cost as per utilization model for easier testing and proof of concepts



Easier licensing option





PostgreSQL in AWS RDS

When you need a feature-rich, open-source relational database system



When you need powerful indexing capabilities and support for complex queries



When you prioritize flexibility and extensibility over ease of use



Atlassian, Blackboard, and Amazon (retail wing) are few of the notable AWS RDS PostgreSQL customers



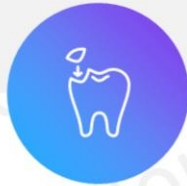


MariaDB in AWS RDS

When you need a feature-rich, open-source relational database system



When you want a drop-in replacement for MySQL



When you value the open-source nature and compatibility with MySQL ecosystem



When you require high availability and scalability options





Microsoft SQL Server in AWS RDS

When you are working in a Windows environment and prefer a Microsoft product



When you need strong integration with other Microsoft technologies and tools



When you require enterprise-level features such as advanced security and high availability



When you have a larger budget and can invest in a commercial database solution



ORACLE



Oracle DB in AWS RDS

When you need
advanced security
features and
comprehensive data
management



When you prioritize
vendor support and
have specific
requirements that
align with Oracle's
offerings



When you are
dealing with crucial
financial data
applications



Customers:
TrueMoney,
Shopdirect, Open
Universities
Australia





Q | MySQL in AWS RDS

When your application or project is based on a LAMP (Linux, Apache, MySQL, PHP/Python/Perl) stack



When you have a smaller budget and need a cost-effective solution



Customers: Airbnb, Freshworks, Flipboard





Choosing the Right Database

Structured
Database

Small microservice serving less traffic



MySQL



MariaDB

Microsoft/Oracle compatible applications

ORACLE

Oracle



Microsoft
SQL Server

High availability, serving major application traffic, extended use
case for machine learning



PostgreSQL



Understanding RDS Licensing

Bring your own license (BYOL) and License included by AWS

ORACLE



Databases like Oracle SQL and Microsoft SQL servers operate with licensing.

Switching allowed between BYOL and AWS licensing

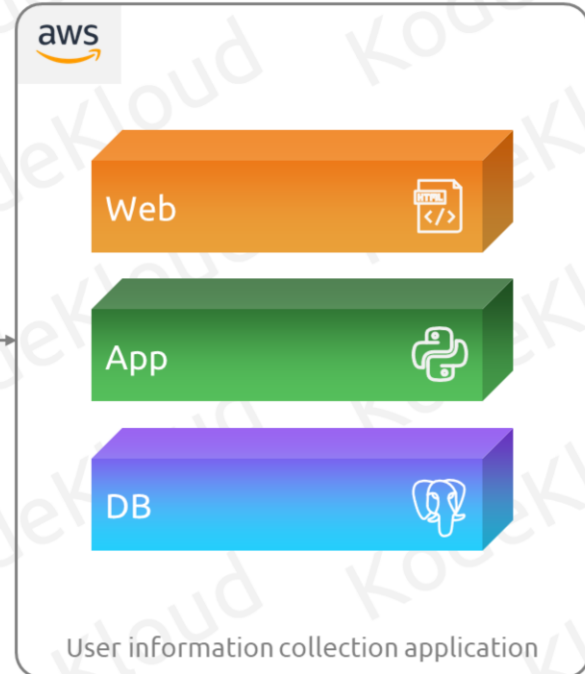
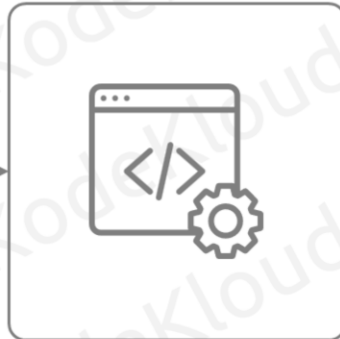
ORACLE



For proprietary database engines like Oracle or SQL Server, you need to bring your own license (BYOL) or use the "license-included" option provided by AWS.

RDS Architecture and Concepts

RDS Architecture and Concepts



Deploying Sample Application on EC2 Instance

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This slide is part of a demo video. I will kick start with a small introduction and then move to AWS console.

Creating and Connecting RDS Instance to Application

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This slide is part of a demo video. I will kick start with a small introduction and then move to AWS console.

RDS Instance Types

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This slide is part of a demo video. I will kick start with a small introduction and then move to AWS console.

RDS Instances Types

General Purpose



Memory Optimized



RDS Instances Types



Balanced Performance

- Good compromise between CPU performance, memory, and network capabilities
- Best suited for applications not receiving heavy traffic loads



Burstable Performance

- Accumulate CPU credits during periods of low utilization
- These credits can be used during spikes in CPU demand

RDS Storage Types

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RDS Storage Types

General Purpose
SSD



Provisioned IOPS
SSD



Magnetic



General Purpose SSD



General Purpose SSD volumes offer **cost-effective** storage



Ideal for a broad range of workloads running on **medium-sized DB** instances



General Purpose storage is best suited for **development and testing** environments



Baseline of **3 IOPS/GB** and ability to burst up to **3,000 IOPS**

Provisioned IOPS SSD



Provisioned IOPS storage is designed to meet the needs of I/O-intensive workloads and backed with **SSD-backed storage**



Particularly suited for database workloads that require **low I/O latency and consistent I/O throughput**



Provisioned IOPS storage is best suited for **production environments**



Minimum storage size is **100GB** and maximum is **16TB**

Magnetic



One of the initial storage types available when RDS was first launched but has been largely replaced by more modern and performant storage options



Magnetic Storage relies on traditional Hard Disk Drives (HDDs) for data storage. HDDs offer slower performance compared to newer storage technologies like Solid State Drives (SSDs)



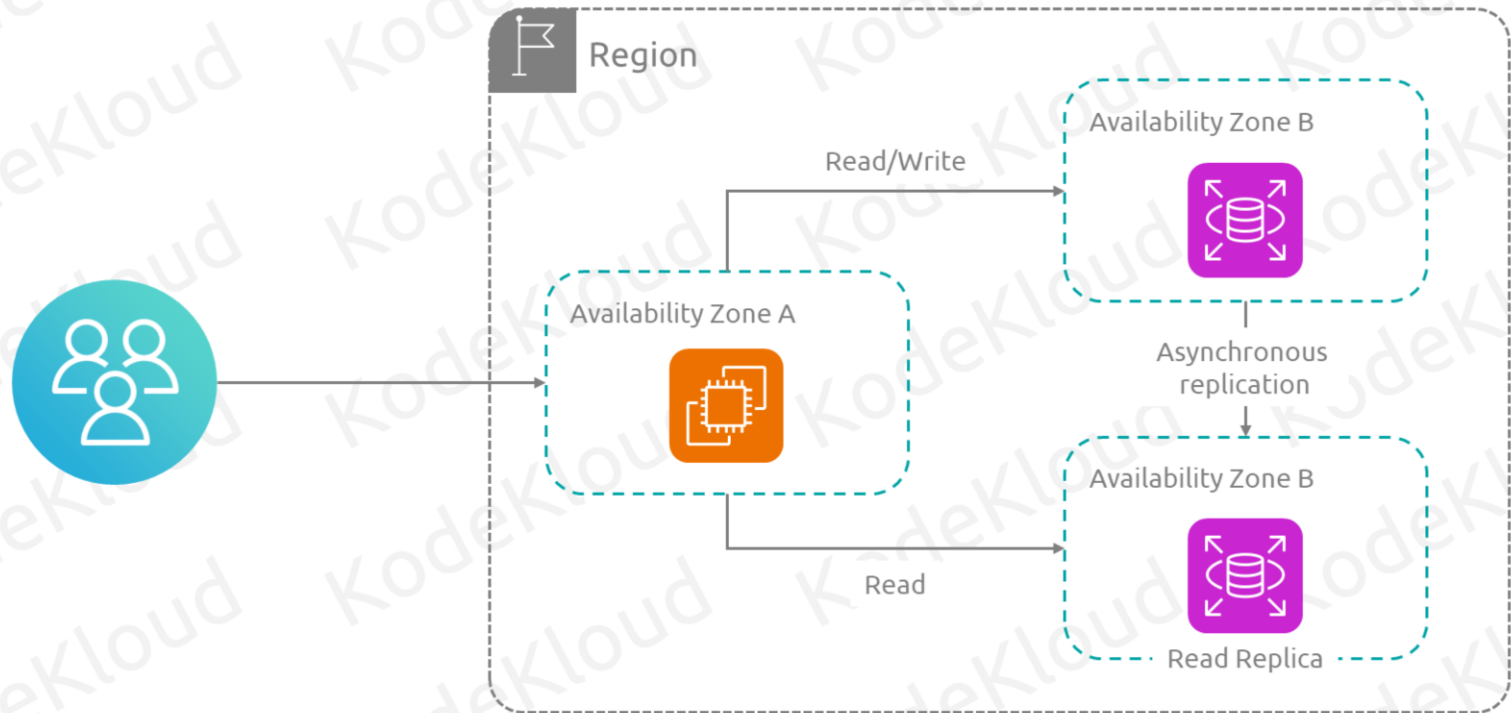
Amazon RDS Magnetic Storage was being phased out and was no longer available for some newer database engine versions and instance types

RDS Read Replicas

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RDS Read Replicas



RDS Read Replicas



A read replica is a read-only copy of a DB instance



Reduce the load on your primary DB instance by routing queries from your applications to the read replica



Elastically scale out beyond the capacity constraints of a single DB instance for read-heavy database workloads



Promote a read replica to a standalone instance as a disaster recovery solution if the primary DB instance fails

Create Read Replicas

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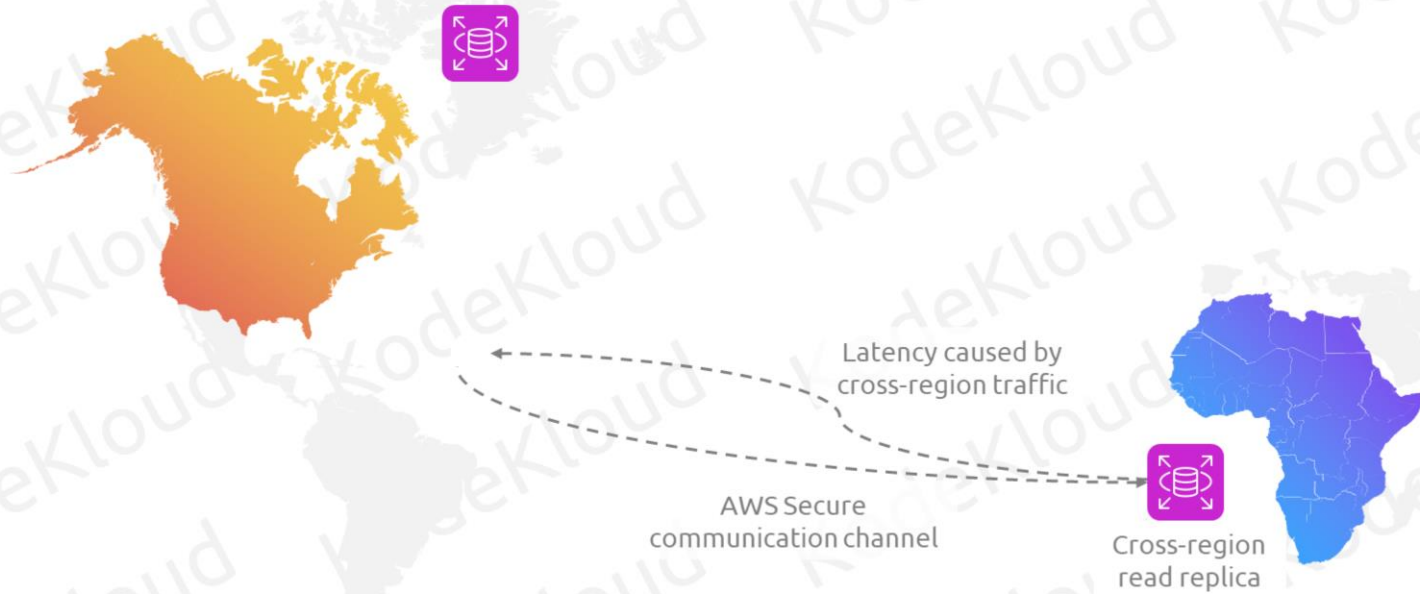
This slide is part of a demo video. I will kick start with a small introduction and then move to AWS console.

RDS Cross-Region Read Replicas

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RDS Cross-Region Read Replicas



RDS Architecture and Core Concepts – Summary

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This slide is part of a demo video. I will kick start with a small introduction and then move to AWS console.

RDS Architecture and Core Concepts – Summary



Practically deployed an application that connects to AWS RDS



Understood the working of AWS RDS instance



Different RDS instance/storage types

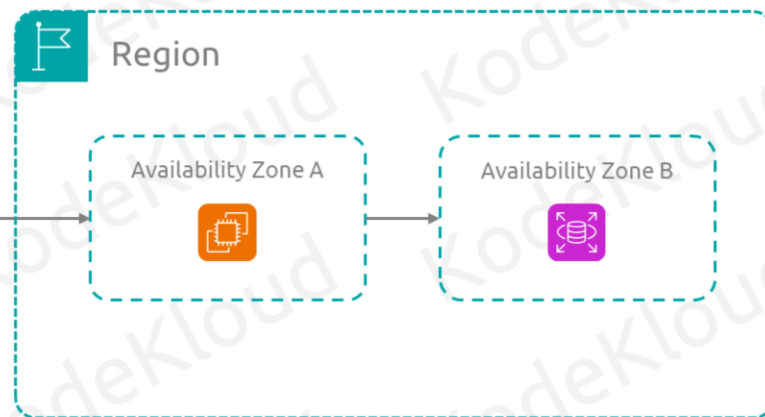


Deep dive into AWS RDS read replicas and cross-region read replicas

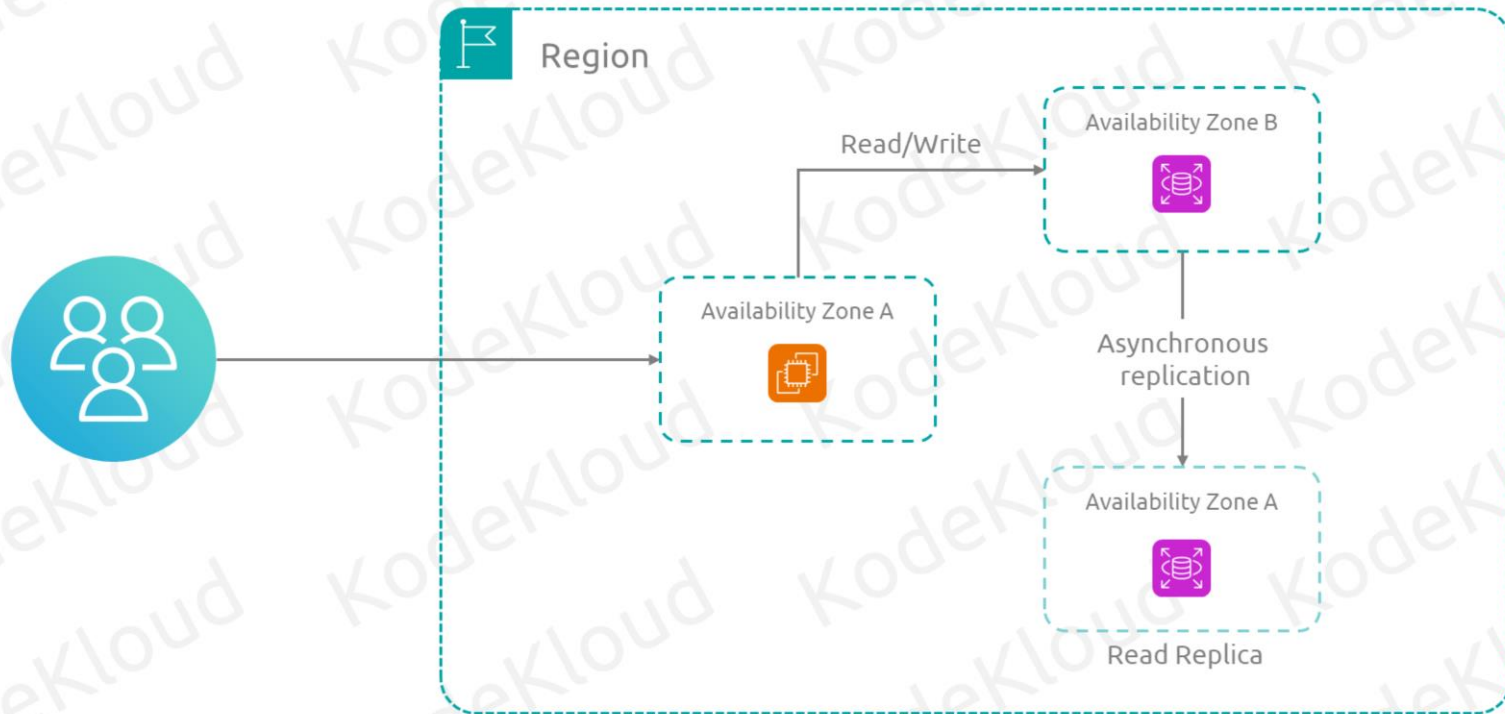
Understanding Single, Multi-AZ Instance and Multi-AZ Cluster setup

Single RDS Database Setup

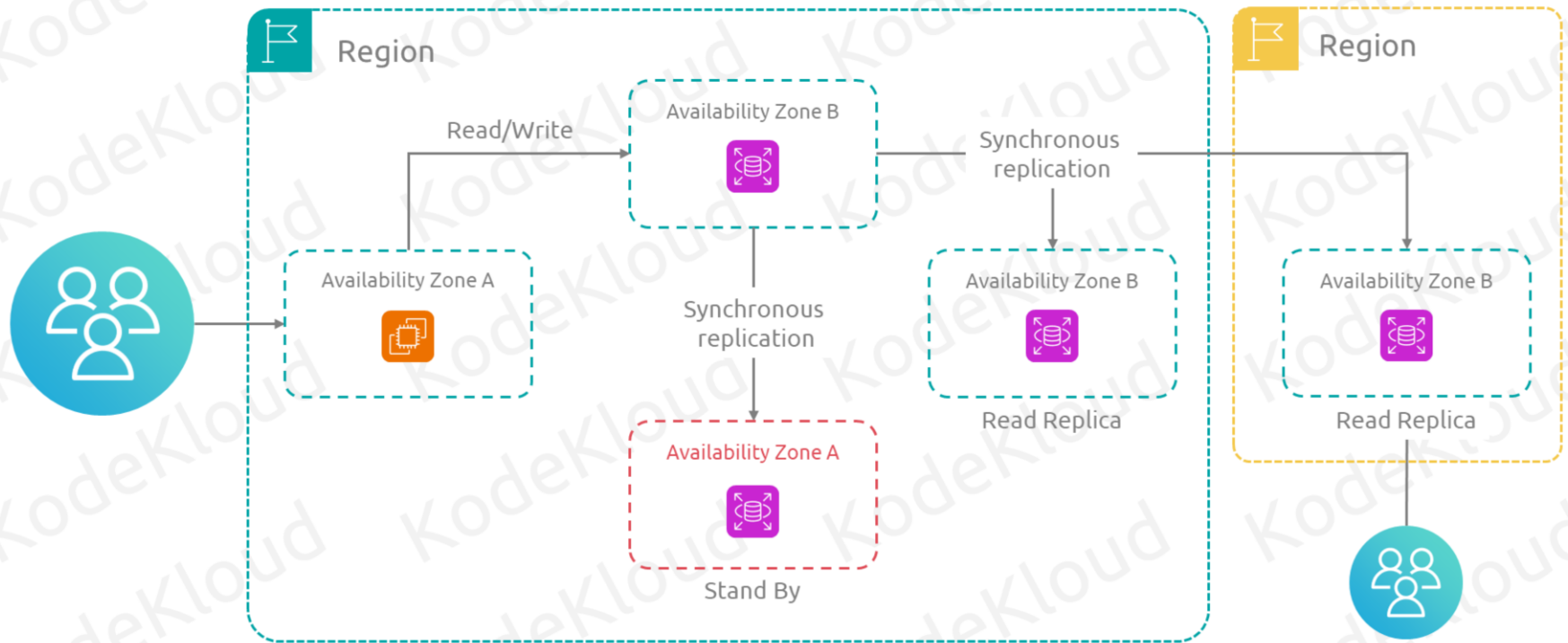
- Best suited for **Staging/Development** environment and running POC projects
- Offers low setup cost
- Doesn't offer high availability
- Offers higher latency for read/write operations



Multi-AZ Instance



Multi-AZ Cluster



Key Points to Note

Choosing the right RDS setup is key to scalability of your application



Multi-AZ setups have an additional **cost factor**



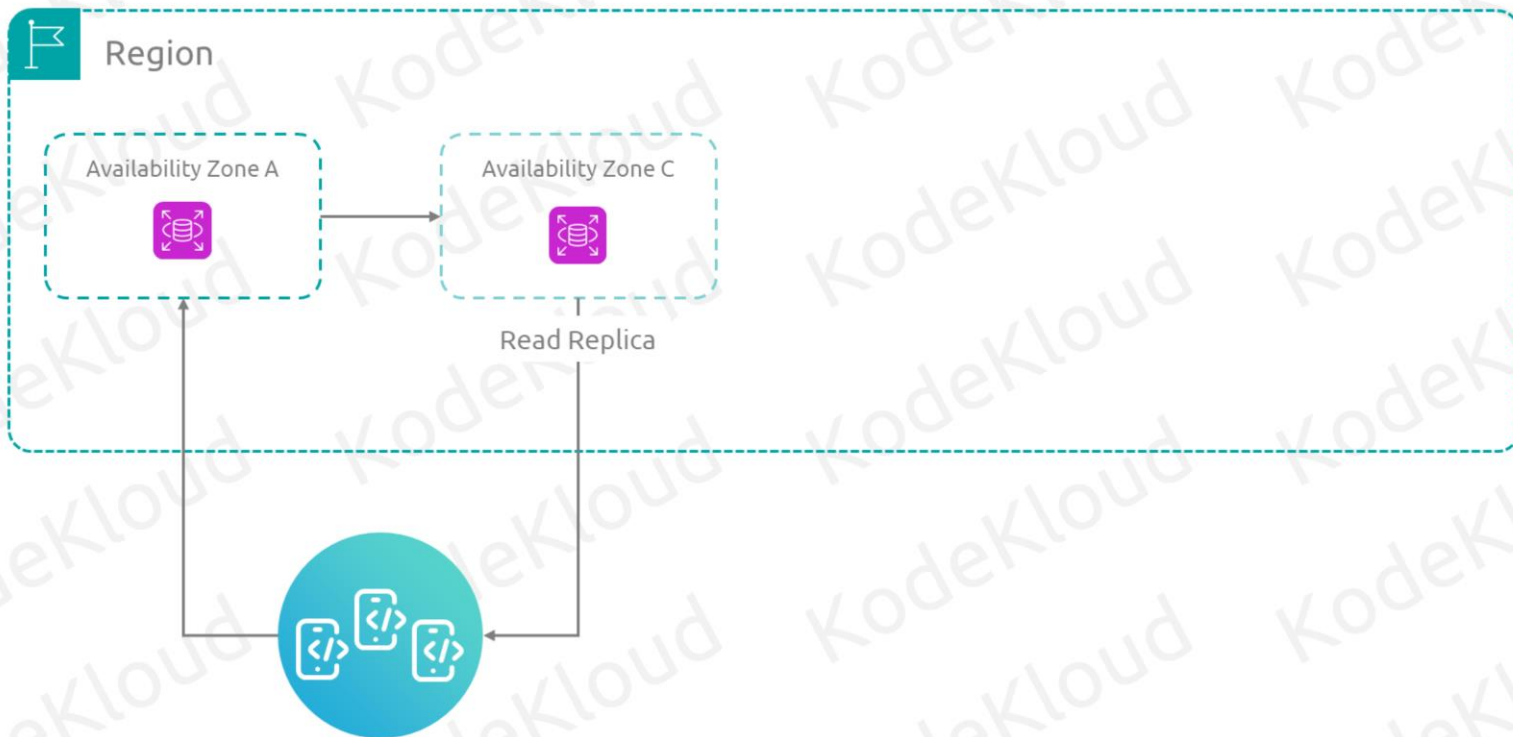
RDS handles data **replication** between instances, ensuring data consistency across the cluster

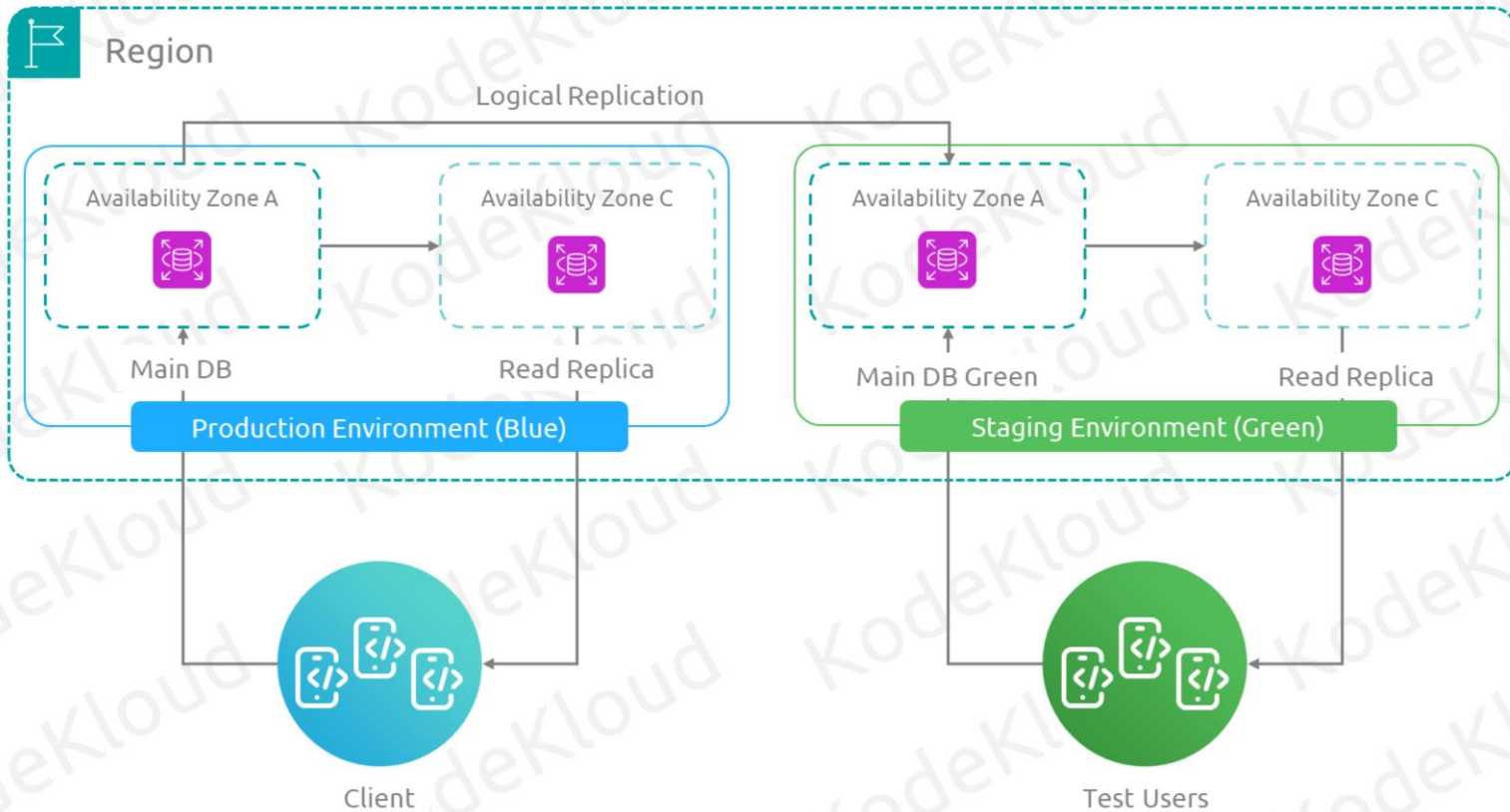


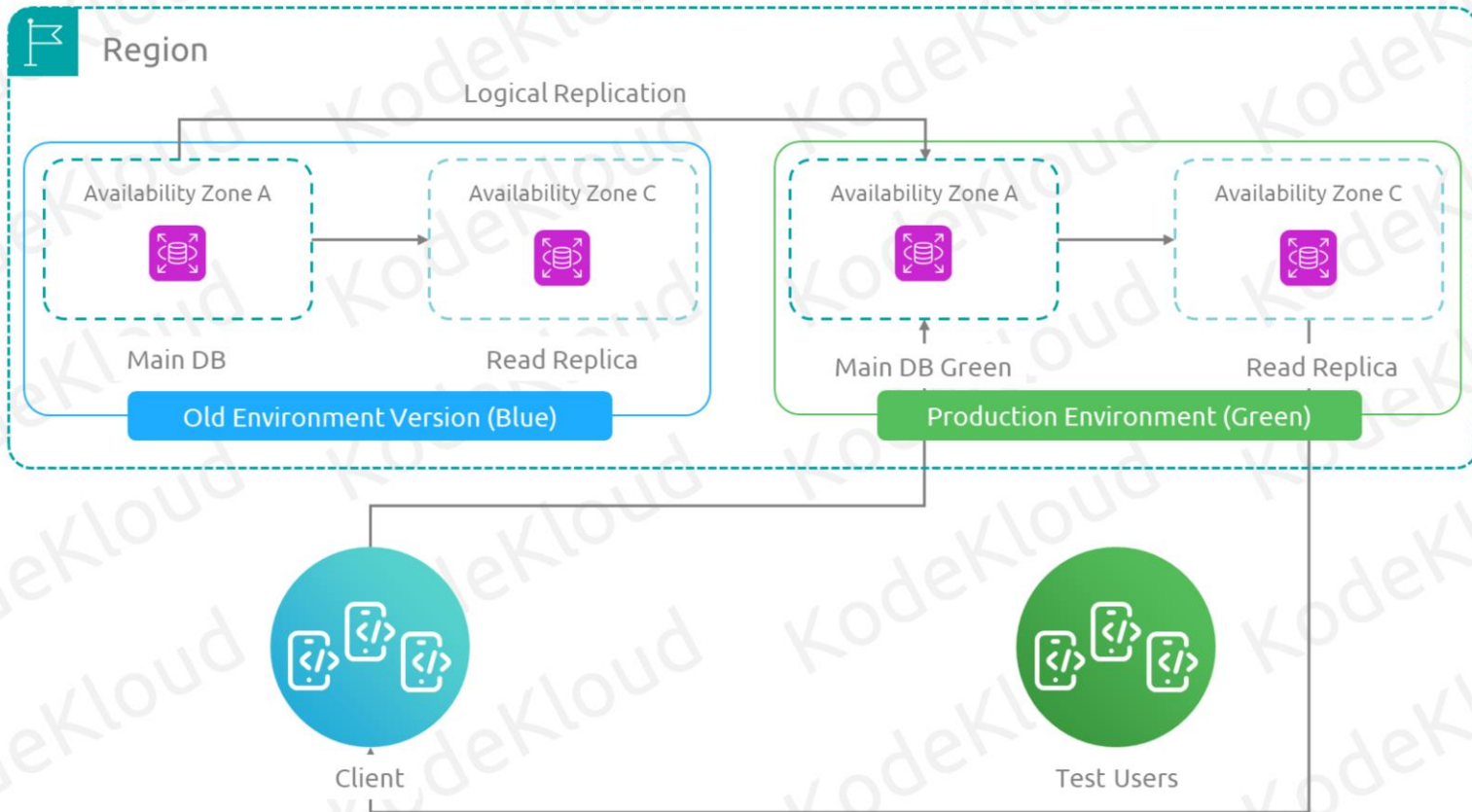
Multi-AZ clusters usually have multiple readers spread across many AZs in the same or a different region



Blue-Green Deployment for Database







Key Points to Note

Test database changes in a safe staging environment without affecting the production environment



The switchover between Blue to Green typically takes under a minute, with no data loss and no need for application changes



The resources in the blue and green environments must be in the same AWS account

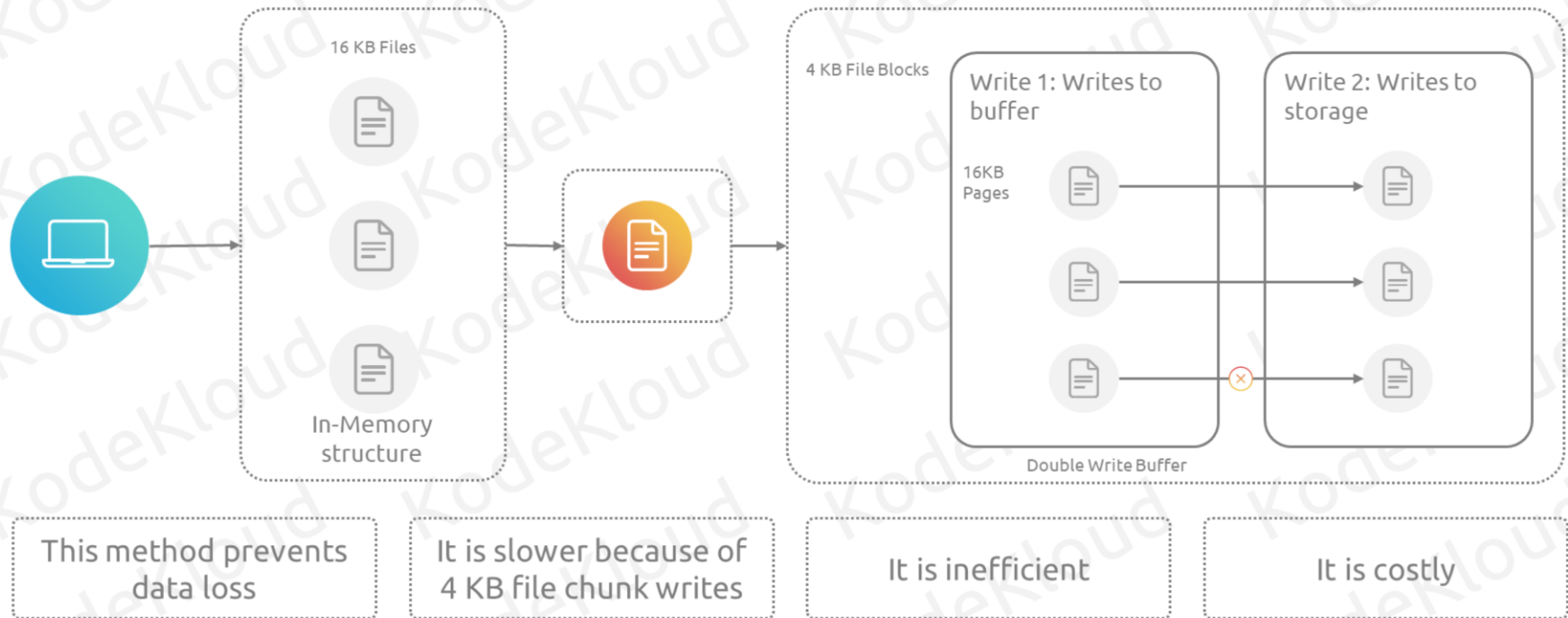


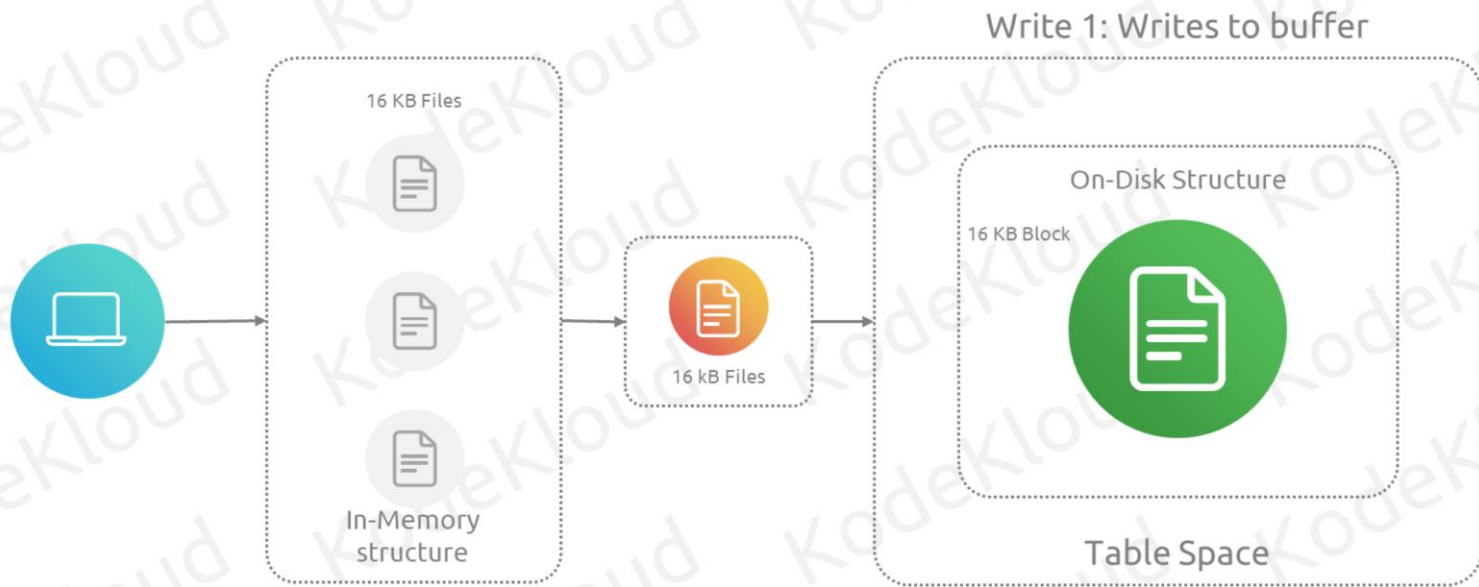
This setup is expensive because two MySQL Clusters are running in parallel



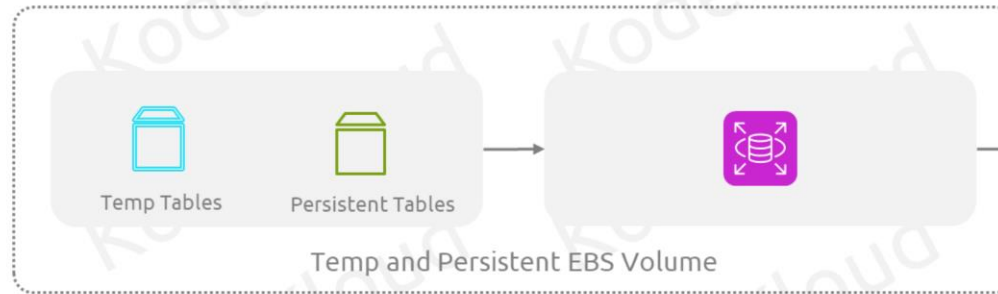
Optimized Read/Write in AWS RDS

MySQL Double Write Buffer

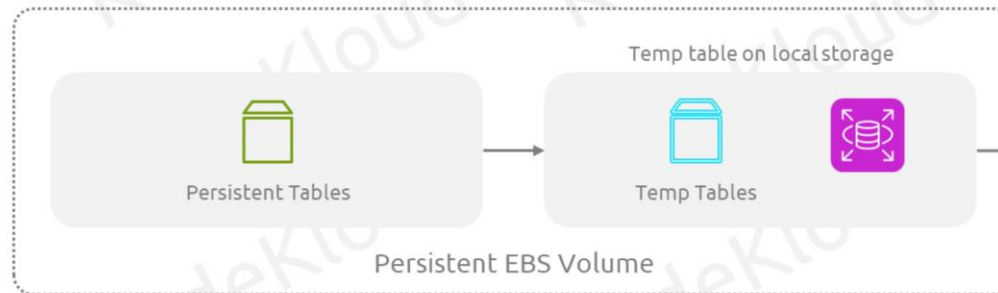




Without optimised reads



With optimised reads



Key Points to Note

With RDS Optimized Reads turned on, your DB instance achieves faster query performance through the use of an instance store



Instance store provides temporary block-level storage for your DB instance



The storage is located on Non-Volatile Memory Express (NVMe) solid state drives (SSDs) that are physically attached to the host server



This storage is optimized for low latency, high random I/O performance, and high sequential read throughput



Understanding RDS Networking and Security

Why Secure Our Database?

Why Secure Our Database?

- 
- User Personal Information
 - Username, Passwords, Tokens
 - Transaction Data

Data Protection



Business Analytics, Machine Learning



Legal Compliance



Continuity and Reliability of Application



1. Data protection: Secure databases safeguard sensitive information from unauthorized access, reducing the risk of data breaches and identity theft.

2. Legal compliance: Implementing robust security measures ensures adherence to data protection laws and industry regulations, mitigating potential legal consequences and penalties.

3. Business reputation: A secure database enhances customer trust, as clients feel confident that their personal information is well-protected from cyber threats.

4. Intellectual property protection: Safeguarding proprietary data and trade secrets is crucial to maintain a competitive advantage in the market.

5. Continuity and reliability: Database security measures prevent data corruption and loss, ensuring uninterrupted business operations and reliable access to critical information.

Understanding RDS Networking and Security

Basics of RDS Networking and Security

RDS Proxy

Database authentication and access management with Amazon RDS



Password management with Amazon RDS and AWS Secrets Manager



Infrastructure security in Amazon RDS



Security best practices for Amazon RDS



Controlling access with security groups



This slide we will be discussing the topics we shall cover in this section



Understanding RDS Networking and Security

Basics of RDS
Networking and Security

RDS Proxy

This slide we will be discussing the topics we shall cover in this section

Database Authentication and Access Management With Amazon RDS

Database Authentication and Access Management With Amazon RDS

Password Authentication



Works well for small application DBs and is the easiest way

IAM Database Authentication



With this authentication method, you use an authentication token to connect to a DB instance; you don't need a password.

Kerberos Authentication



External authentication of database users using Kerberos and Microsoft Active Directory

Database Authentication and Access Management With Amazon RDS

Service User



Engineers who need access to DB to perform minor tasks like query data, etc.

Service Administrator



You're in charge of Amazon RDS resources for your project. You can change RDS instance types, networking, and other activities

Administrator



The security team of your organization who monitor ingress/egress traffic and monitor VPC, Security group, and IAM policies

Service user - Engineers who need access to DB to perform minor tasks like query data, etc.

Service administrator - You're in charge of Amazon RDS resources for your project, You can change RDS instance types, networking and other activities

Administrator - Usually security team of your organization who monitor ingress/egress traffic and monitor VPC, Security group and IAM policies

Password Management With Amazon RDS and AWS Secrets Manager

Password Management With Amazon RDS and AWS Secrets Manager

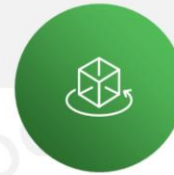
RDS automatically generates database credentials



Stores RDS database credentials in AWS Secrets Manager directly; other applications can use it from a single controlled source



Rotates database credentials regularly, without requiring application changes



Secrets Manager allows fine-grained control of access to database credentials in secrets using IAM



RDS automatically generates database credentials.

RDS automatically stores and manages database credentials in AWS Secrets Manager.

RDS rotates database credentials regularly, without requiring application changes.

Secrets Manager allows fine-grained control of access to database credentials in secrets using IAM.

Password Management With Amazon RDS and AWS Secrets Manager – Demo

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RDS automatically generates database credentials.

RDS automatically stores and manages database credentials in AWS Secrets Manager.

RDS rotates database credentials regularly, without requiring application changes.

Secrets Manager allows fine-grained control of access to database credentials in secrets using IAM.

Infrastructure Security in Amazon RDS

Infrastructure Security in Amazon RDS

As a managed service, Amazon Relational Database Service is protected by AWS global network security



AWS Shared responsibility model



Make sure you set up RDS based on the application usage, understanding the AWS Global network



As a managed service, Amazon Relational Database Service is protected by AWS global network security
Share responsibility model
Make sure you setup RDS based on the application usage, understanding the AWS Global network

Security Best Practices for Amazon RDS

Security Best Practices for Amazon RDS – Summary

Create an individual user for each person who manages Amazon RDS resources, including yourself. Don't use AWS root credentials to manage Amazon RDS resources.

Grant each user the minimum set of permissions required to perform his or her duties.

Use IAM groups to effectively manage permissions for multiple users.

Rotate your IAM credentials regularly.

Configure AWS Secrets Manager to automatically rotate the secrets for Amazon RDS.

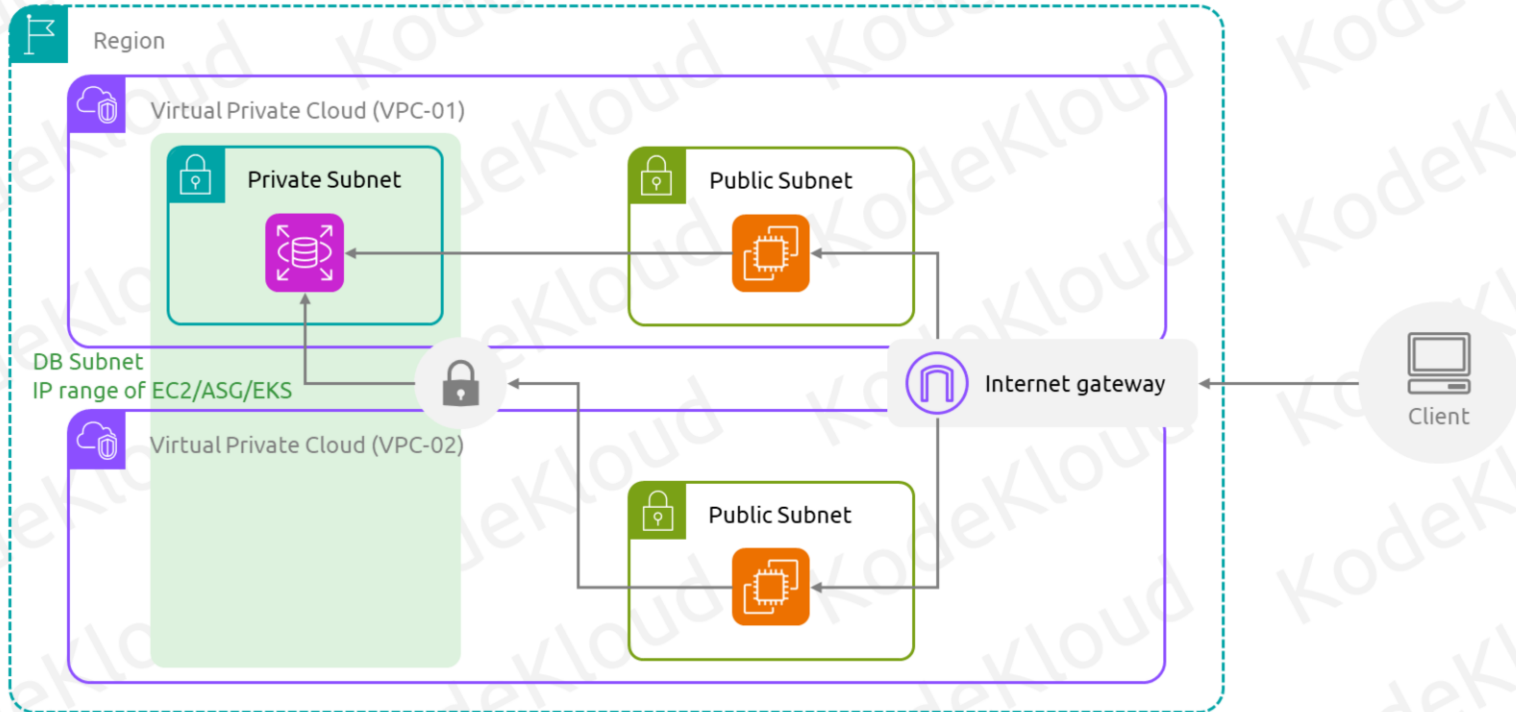
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- Create an individual user for each person who manages Amazon RDS resources, including yourself. Don't use AWS root credentials to manage Amazon RDS resources.
- Grant each user the minimum set of permissions required to perform his or her duties.
- Use IAM groups to effectively manage permissions for multiple users.
- Rotate your IAM credentials regularly.
- Configure AWS Secrets Manager to automatically rotate the secrets for Amazon RDS. For more information,

see [Rotating your AWS Secrets Manager secrets](#) in the *AWS Secrets Manager User Guide*. You can also retrieve the credential from AWS Secrets Manager programmatically. For more information, see [Retrieving the secret value](#) in the *AWS Secrets Manager User Guide*.

Controlling Access With Security Groups

Controlling Access With Security Groups



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Show the screenshot of Public access / Internal Access from RDS creation homepage
Understand security groups and how it helps us control access – Diagram

Controlling Access With Security Groups

Security groups are the first line of defense for your database.

VPC security groups control the access that traffic has in and out of a DB instance.

By default, network access is turned off for a DB instance.

You can specify rules in a security group that allow access from an IP address range, port, or security group.

You can specify up to 20 rules in a security group.

VPC security groups control the access that traffic has in and out of a DB instance

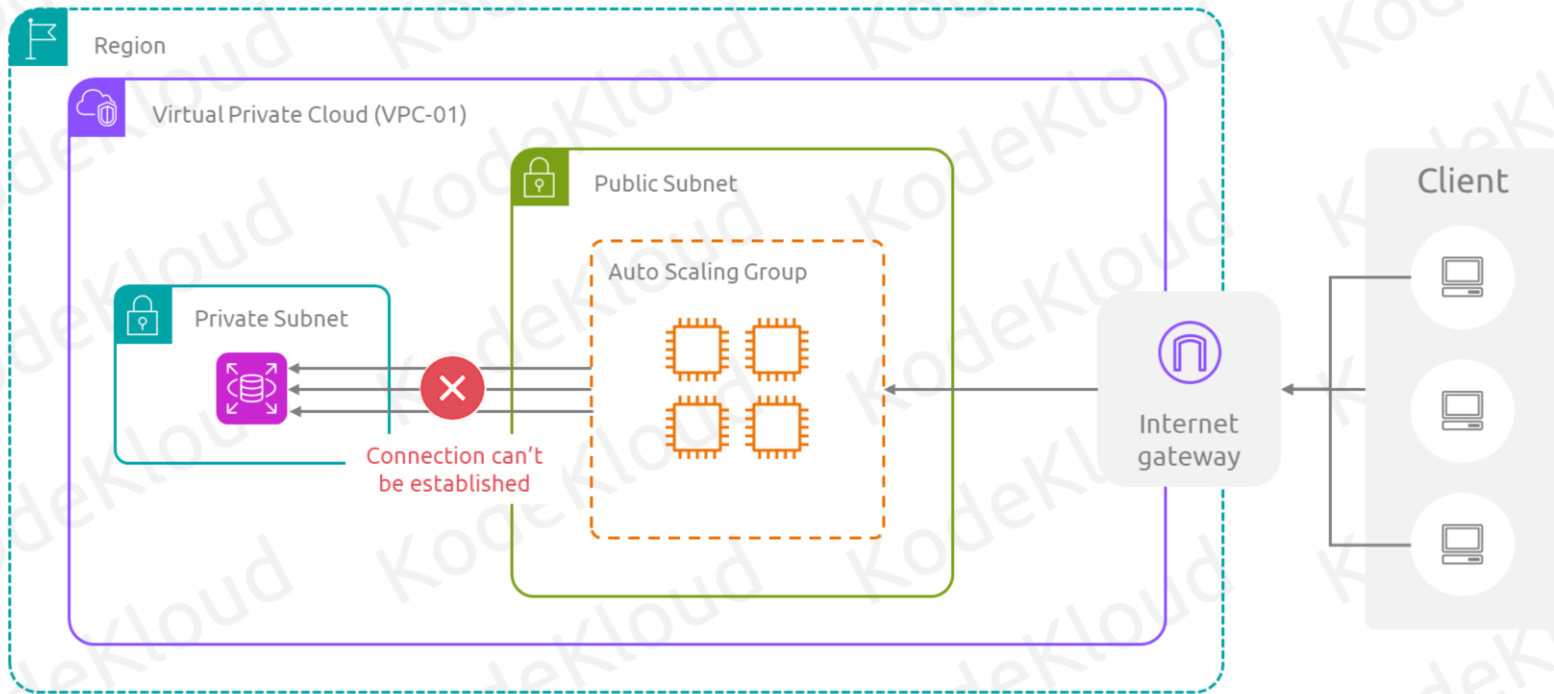
By default, network access is turned off for a DB instance

You can specify rules in a security group that allow access from an IP address range, port, or security group

You can specify up to 20 rules in a security group.

AWS RDS Proxy

AWS RDS Proxy



AWS RDS Proxy



Amazon RDS Proxy improves application scalability by pooling and sharing database connections.

AWS RDS Proxy



It makes applications more resilient to database failures by automatically connecting to a standby DB instance while preserving application connections.

AWS RDS Proxy

RDS Proxy allows you to enforce IAM authentication for databases and securely store credentials in AWS Secrets Manager.

AWS RDS Proxy



The connection pool approach avoids the memory and CPU overhead of opening a new database connection each time.

AWS RDS Proxy

RDS Proxy queues or throttles application connections that can't be served immediately from the pool of connections, maintaining predictable performance for the load that can be served.

AWS RDS Proxy

RDS Proxy reduces the overhead to process credentials and establish a secure connection for each new connection, which can be handled on behalf of the database.

Working With Amazon RDS Proxy Endpoints – Demo

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VPC security groups control the access that traffic has in and out of a DB instance

By default, network access is turned off for a DB instance

You can specify rules in a security group that allow access from an IP address range, port, or security group

You can specify up to 20 rules in a security group.

Backup and Restore With RDS

Backup and Restore With RDS

Why do we need a backup of database?



Disaster Recovery

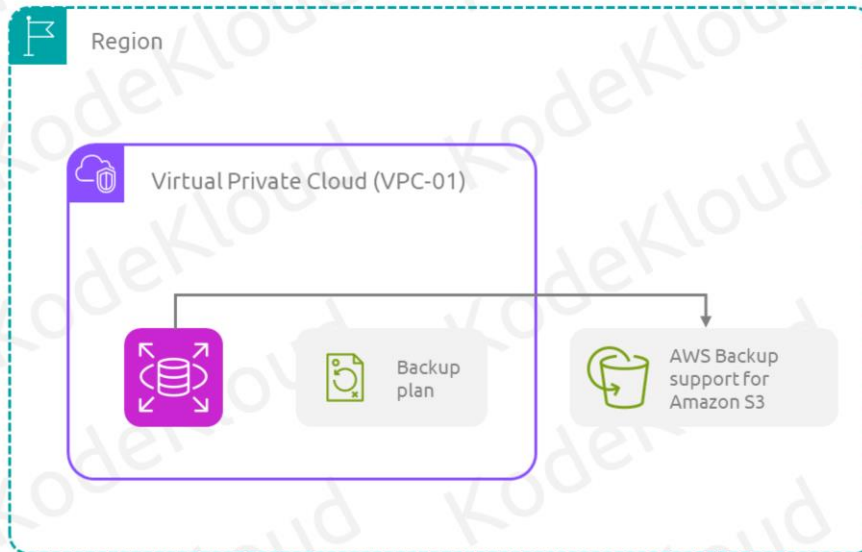


Data Loss Prevention



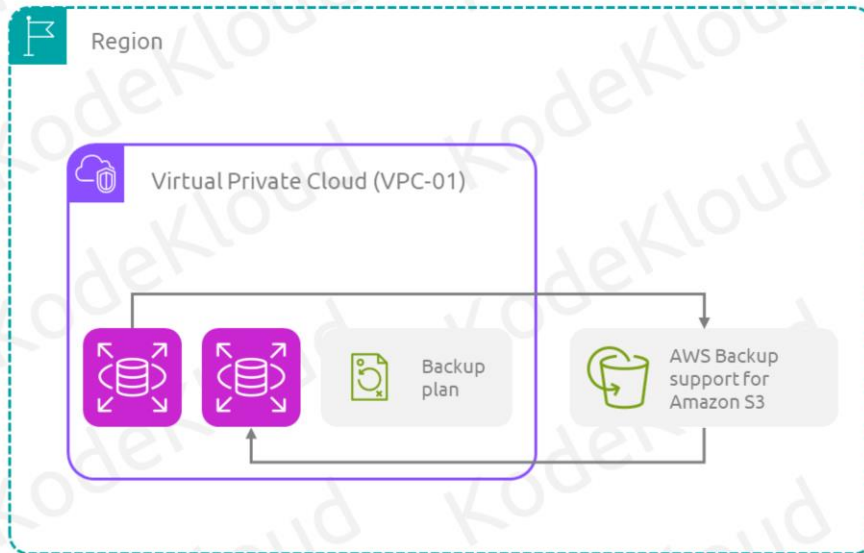
Compliance

Backup and Restore With RDS



- Backup storage for each AWS Region is composed of the automated backups and manual DB snapshots for that Region
- Total backup storage space equals the sum of the storage for all backups in that Region
- Backups are stored in Amazon S3
- Automated backups occur daily during the preferred backup window
- Automated backups are not created when a DB instance or cluster is stopped

Backup and Restore With RDS

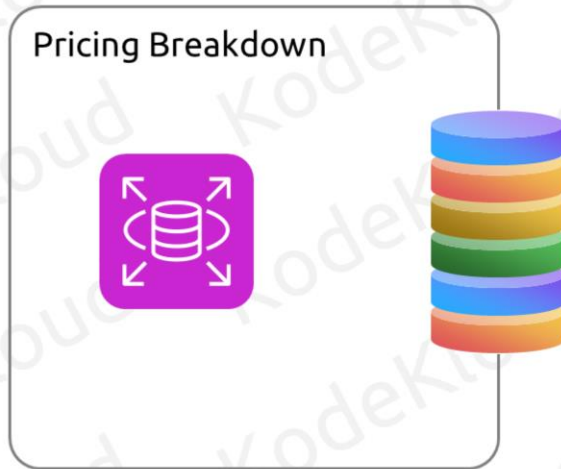


- Point-in-time backup
- If you disable and then re-enable automated backups, you can restore starting only from the time you re-enabled automated backups
- We can restore a DB in a new region if required

Backup and Restore With RDS Lab

Costing for Your RDS Instance

Costing for Your RDS Instance



Instance type

I/O request per month

Provisioned IOPS

Multi-AZ deployment

Storage

Data transfer

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- DB instance hours – Based on the type (e.g. db.t2.micro, db.m4.large) of the DB instance consumed. Following a billable status change, such as creating, starting, or modifying your DB instance type, you will be billed based on partial DB instance hours. Partial DB instance hours consumed are billed in one-second increments with a 10-minute minimum. For additional details, read our [what's new announcement](#).

- Storage (per GB per month) – [Storage capacity](#) you have provisioned to your DB instance. If you scale your provisioned storage capacity within the month, your bill will be pro-rated.

- I/O requests per month – Total number of [storage I/O requests](#) (for Amazon RDS magnetic storage and Amazon Aurora only).
- Provisioned IOPS per month – Provisioned IOPS rate, regardless of IOPS consumed (for Amazon RDS Provisioned IOPS storage only).
- Backup Storage – The storage associated with your automated database backups and any customer-initiated database snapshots. Increasing your backup retention period or taking additional database snapshots increases the backup storage consumed by your database.
- Data transfer – [Internet data transfer in and out of your DB instance](#).

Costing for Your RDS Instance

Pricing breakdown for an AWS RDS multi-zone cluster

Monthly Cost Breakdown

Database instances
(2 x db.r5.large)

\$1,318.40
(\$659.20 per instance)

Multi-AZ deployment

\$1,290
(\$0.15 per GB-month of
provisioned storage; for
8,600 GB)

Backup storage

\$10
(\$0.20 per GB-month; for 50
GB)

Data transfer

\$0
(Free within the same
region)

Total monthly cost

\$2,618.40

Running a db.r5.large instance in each Availability Zone (AZ)

Running in the US East (N. Virginia) region

500 GB of storage per instance

Running PostgreSQL database engine

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Here's an example pricing breakdown for an AWS RDS multi-zone cluster:

Assumptions:

- Running a db.t3.small instance in each Availability Zone (AZ)
- 30GB of storage per instance
- Running in the US East (N. Virginia) region
- Running PostgreSQL database engine

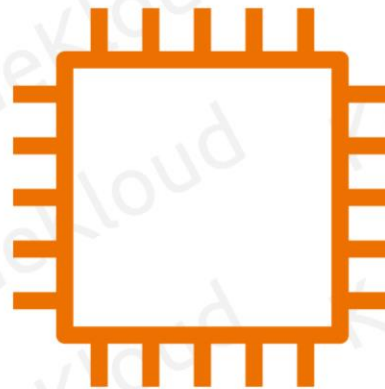
Monthly cost breakdown:

- Database instances (2 x db.t3.small): \$54.88 (\$27.44 per instance)
- Multi-AZ deployment: \$44.64 (\$0.15 per GB-month of provisioned storage)
- Backup storage: \$2.40 (\$0.20 per GB-month)
- Data transfer: \$0 (free within the same region)
- Total monthly cost: \$101.92

Costing for Your RDS Instance – Pitfalls to Avoid



Choose the right instance type for your workload



If you choose an instance type that is too large for your needs, you will be paying for **resources** you don't use. If you choose an instance type that is too small, your database may not perform well.

Costing for Your RDS Instance – Pitfalls to Avoid



Choose the right instance type for your workload



If you choose an instance type that is too large for your needs, you will be paying for resources you don't use. If you choose an instance type that is too small, your **database** may not perform well.

Costing for Your RDS Instance – Pitfalls to Avoid



Monitor your usage

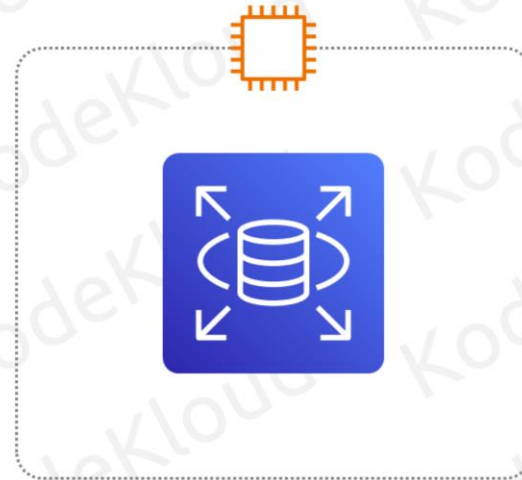


Use **Amazon CloudWatch** to monitor your database usage and make sure you are not exceeding your allotted resources.

Costing for Your RDS Instance – Pitfalls to Avoid



Use reserved instances



If you plan to use **RDS** for an extended period of time, consider purchasing a reserved instance. This can save you up to **75% compared** to on-demand pricing.

Costing for Your RDS Instance – Pitfalls to Avoid



Make sure your **database** is optimized for performance and that you are not wasting resources on unnecessary operations.



Optimize your database

What Is AWS Aurora?

What Is AWS Aurora?

- Enterprise database at open-source pricing
- Delivered as a managed service



Features



Speed and availability
of high-end
commercial databases



Drop-in compatibility
with MySQL and
PostgreSQL



Simple pay-as-you-go
pricing



Fully managed service,
automating
administrative tasks

Aurora Architecture vs RDS – What's the Difference?

01

Storage Architecture

- Unlike RDS, which uses standard storage, Aurora uses a distributed storage architecture
- It replicates data across multiple nodes, providing better performance and fault tolerance

02

03

04

05

01

Performance

02

- Aurora's architecture makes it more efficient for read-heavy workloads
- It offloads reads to replicas, reducing the load on the primary instance

03

04

05

01

Replication

02

- Aurora has a quorum-based approach, which allows for faster crash recovery and less replication lag

03

04

05

01

Automatic Failover

02

- Aurora provides automated failover with zero data loss, while RDS may experience some data loss during failover

03

04

05

01

Backup and Restore

02

- Aurora backups are continuous and incremental, reducing backup times and improving restore performance compared to RDS

03

04

05

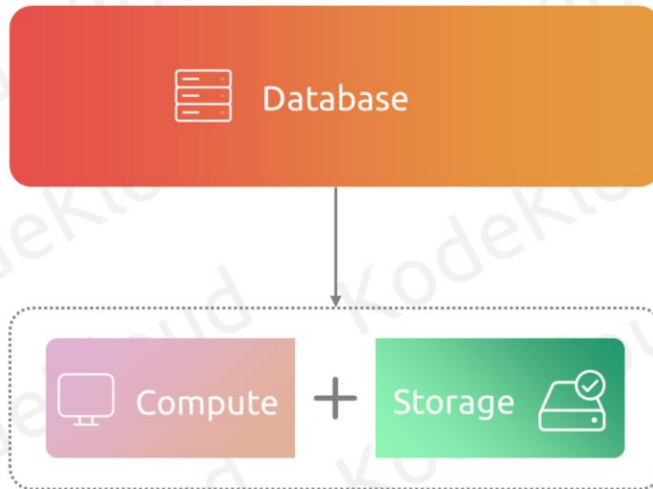
Setting up and Managing Aurora Demo

Aurora DB – How to Set It up Easily

Lab

AWS Aurora – Distributed Storage

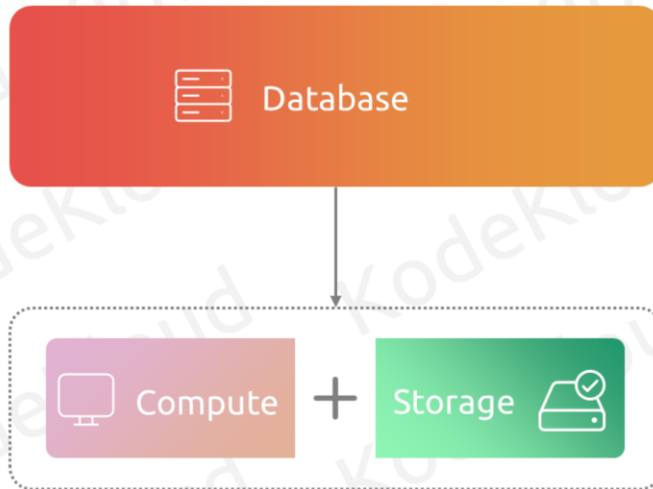
Traditional Database Architecture



Compute and storage have different lifetimes

- ✓ Instances fail and may be replaced
- ✓ Instances are shut down
- ✓ Instances are scaled up/down
- ✓ Instances are added to cluster to scale out

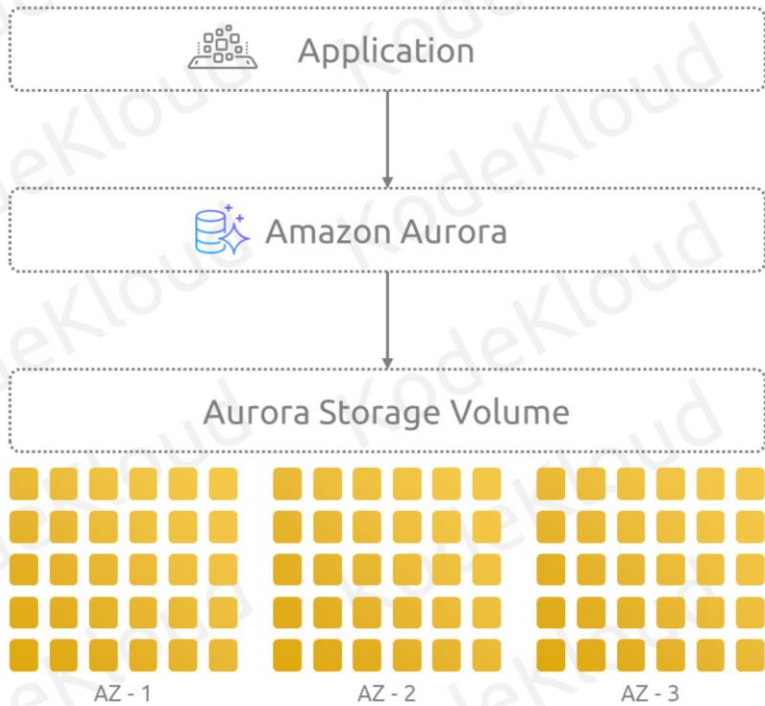
Traditional Database Architecture

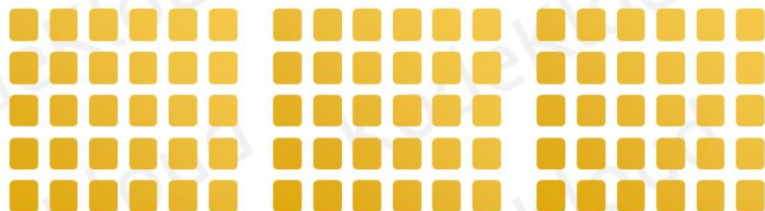
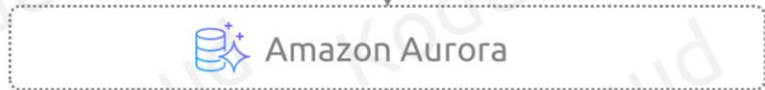
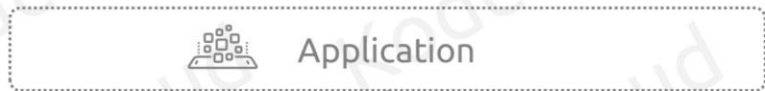


A database is typically a combination of compute and storage in a single instance

Compute and storage are best decoupled for

- Scalability
- Availability
- Durability





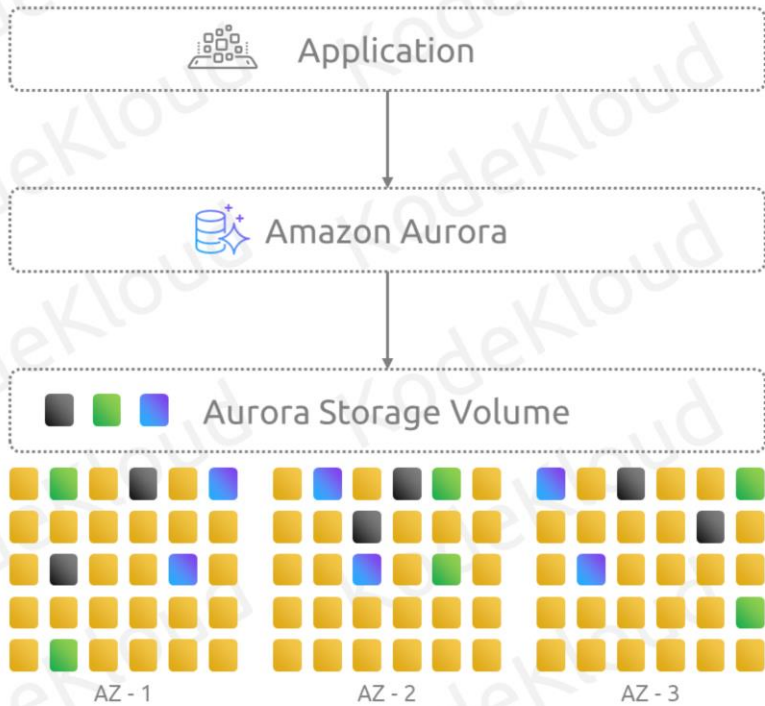
AZ - 1

AZ - 2

AZ - 3



Storage volume
segmented in **10-GB**
protection groups (PG)

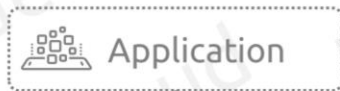


Storage volume segmented in **10-GB** protection groups (PG)

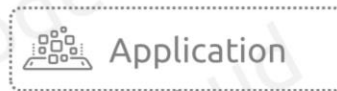
Each PG contains **six 10-GB segments** and copies of the same data on different storage nodes, **two in each AZ**

Six copies of data, two in each AZ to protect against **AZ+1** failure modes

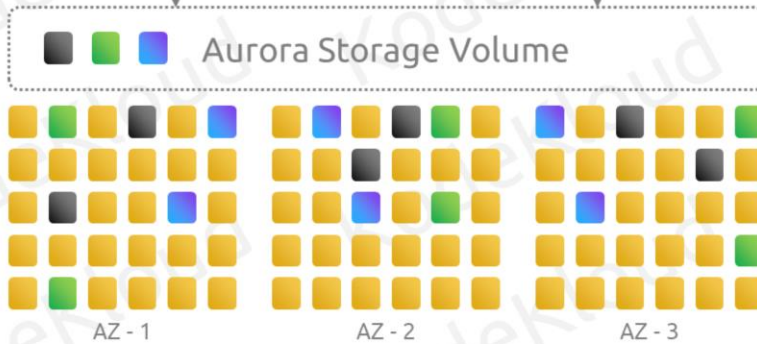
Storage volume grows automatically by adding PGs, up to **128 TB**



Write



Read



Applications will have an easier **write/read I/O**

Using quorum model for writes and reads:

- 4 out of 6 for writes
- 3 out of 6 for reads (recovery)

Maintains write capability if an **AZ fails**; maintains read capability if **AZ + 1** storage node fails

Uses “**gossip protocol**” for node block repair

Backup and Recovery With Aurora



Continuous Backup

Aurora performs continuous backups, capturing changes at the transaction level and minimizing the risk of data loss.



Point-in-Time Restore

Aurora allows you to restore the database to any specific point in time within the backup retention period.



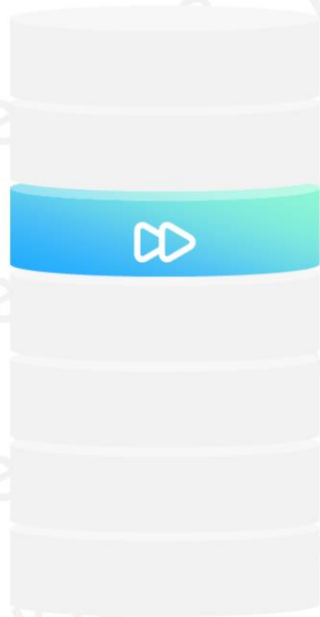
Snapshot Creation

You can manually trigger snapshots of your Aurora database, which can be used for disaster recovery or migrating data to another region.



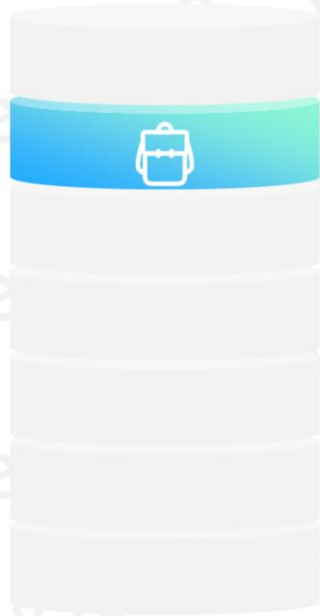
Backup Encryption

Aurora supports encryption for both backups and the underlying storage, enhancing data security.



Fast Restore

Aurora's backups are incremental, enabling faster restore times compared to traditional database backups.



Backtrack Feature

The backtrack feature allows you to move the database back to a previous state without needing to restore from backups.

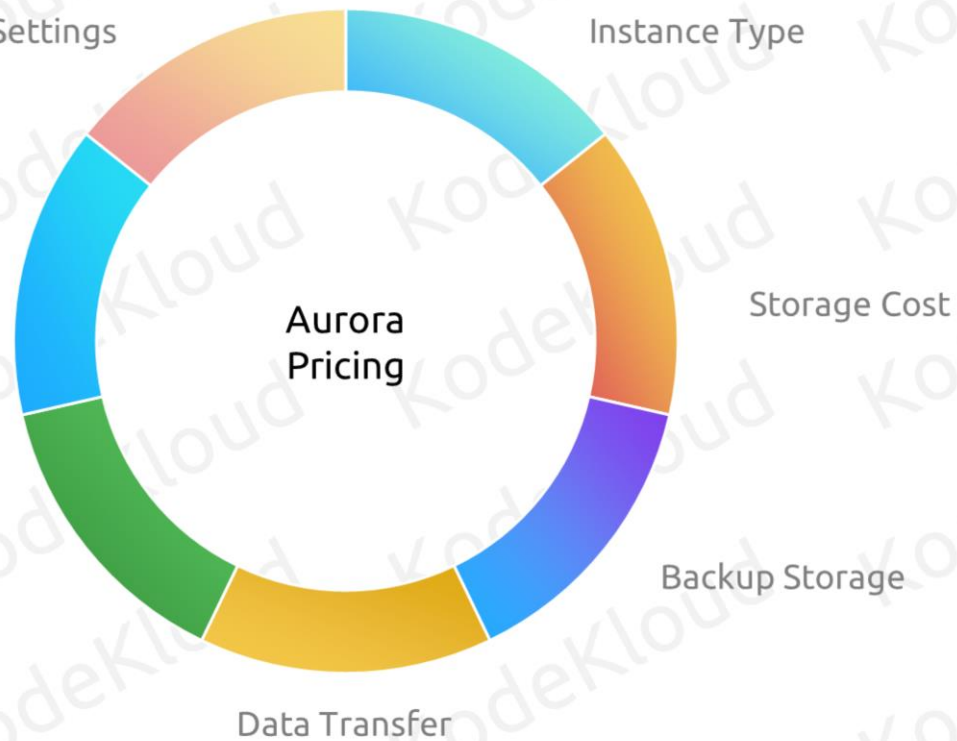


Cross-Region Copy

You can copy Aurora snapshots to different AWS regions, ensuring disaster recovery and reducing data access latency for global applications.

Backup and Recovery With Aurora Demo

Aurora – Pricing



Instance Type

db.r5.large (2 vCPUs, 16 GB RAM)

\$144 per month

Storage

500 GB provisioned storage

\$50 per month

Backup Retention

7 days

\$6 per month

Estimated I/O Operations

10,000 read IOPS and 5,000 write IOPS per day

\$1.50 per month

Data Transfer

500 GB data transferred in, and
200 GB data transferred out per month

Data transfer in cost

500 GB * \$0.10/GB = \$50

Data transfer out cost

200 GB * \$0.15/GB = \$30

Multi-AZ Deployment

Enabled for high availability

\$28.80

Total cost

\$310.13 per month

Aurora vs Aurora Serverless V2



Aurora

- Autoscaling configuration
- Helps adjust the capacity for your databases
- You're charged only for the resources that your DB clusters consume
- Projects planned with no clear capacity



Aurora Serverless V2

- On-demand
- Helps automate the processes of workload monitoring
- Capacity is adjusted automatically based on application demand
- Best suited for variable workloads

AWS RDS – Alerting and Monitoring

Importance of Monitoring Database Performance and Availability



Proactive Issue
Detection



Optimal Performance



Availability and
Reliability



Capacity Planning and
Scaling

How can I monitor AWS RDS ?



Set up CloudWatch
integrations to monitor
key RDS metrics



AWS RDS performance
insights



AWS RDS long-running
query logs



Adapting AWS RDS
recommendations

CloudWatch and RDS Integrations Demo

Performance Insights Demo

Query Editor Console Demo



KodeKloud